YASSINE ABASSI

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SUMMARY

Computer engineering student at Polytechnique Montréal with hands-on experience building full-stack data platforms and real-time systems. Strong foundation in Python, Angular, .NET, and data pipeline design. Passionate about applying computer engineering to solve complex real-world challenges.

EDUCATION

B.Eng. Computer Engineering, Polytechnique Montréal — Aug 2022 – May 2026
120 credits (105 completed, 15 in progress)

EXPERIENCE

Software Quality Assurance Intern — May 2024 – Sep 2024
UPTOTEST · Ariana, Tunisia

- Designed and automated end-to-end functional test suites for the KOORS web app using Cypress with Cucumber/Gherkin in a BDD workflow.
- Translated product requirements into testable user stories and acceptance criteria, managed defects and traceability in Jira across sprints.
- Participated in Agile Scrum ceremonies to align QA priorities with development milestones and release timelines.

TECHNICAL SKILLS

Languages	C, C++, C#, Java, Kotlin, Swift, Python, JavaScript, TypeScript, VHDL, Assembly
Web & Mobile	Angular, React.js, Node.js, NestJS, Express.js, WebSockets, SignalR, SwiftUI, Jetpack Compose, .NET, Three.js
Data & APIs	PostgreSQL, MongoDB, SQLite, Room, Firebase, Retrofit, OkHttp, Entity Framework Core, GDAL, RxJS, Upstash Redis
DevOps & Tools	Git, GitLab, Docker, Jira, Swagger, Qt, Cypress, Jest, Jasmine, Mocha, Vivado, Wireshark
Robots & Boards	Zynq7000, ATmega164A, Pynq-Z1, Basys3 Artix-7, AgileX Limo

PROJECTS

Hydro-Québec Weather Forecasting Platform | *Angular · .NET · EF Core · GDAL · Leaflet · SignalR · Docker*

- Built a full-stack weather visualization platform with Angular and ASP.NET Core for multi-model weather display with interactive geospatial mapping and real-time updates via SignalR.
- Implemented GDAL raster processing, Leaflet cartographic rendering; designed RESTful APIs alongside an SQLite database and containerized the application with Docker to streamline development and deployment.

Android Mobile Applications (3 apps) | *Kotlin · Jetpack Compose · Room · Google Maps SDK · Retrofit*

- Finance tracker with MVVM, Room persistence, and Material 3 Adaptive layouts; Bluetooth device tracker with GPS trajectories, Maps SDK, and Coroutines; location-based running-walking app with Retrofit/OkHttp networking, Firebase cloud synchronization, and Compose Navigation.

School Tracker iOS Application | *Swift · SwiftUI · Combine · UserDefaults*

- iOS app in SwiftUI for tracking classes and coursework with weighted progress and grade calculations, calendar-based deadline planning aggregating due dates across all classes, and on-device persistence via UserDefaults with Codable JSON encoding.

Multi-Robot Exploration System | *Python · ROS2 · Gazebo · Docker*

- Designed a multi-robot coordinated exploration system with AgileX Limo robots, implementing navigation, and sensor fusion in containerized modules.

Real-Time Multiplayer Combat Game | *Angular · NestJS · WebSockets*

- Developed a real-time multiplayer game with bidirectional WebSocket communication, matchmaking, event synchronization, and responsive UI.

Desktop Chess Application | *C++ · Qt* | **Github ↗**

- Built a fully interactive chess application implementing legal move validation for all piece types with a custom QWidget board renderer.

Embedded Obstacle Detection System | *C++ · ATmega164A*

- Built an autonomous obstacle detection and search system with low-level I/O control, and real-time decision logic on the ATmega164A microcontroller.

Lost Woods Browser Horror Game | *React.js · Typescript · Web Audio API* | **Live Demo ↗**

- Built a full-featured canvas-based horror game with procedural forest tile generation, and reachable key placement using pathfinding logic.
- Engineered enemy AI behaviour, flashlight-reactive threat logic, stamina system, purge spell with cooldown, and separate indoor/outdoor scenes.

3D Interactive Portfolio | *React.js · Three.js · Upstash Redis* | **Live Demo ↗**

- Crafted a 3D portfolio with an interactive room featuring a functional OS, custom camera rigging, and state-driven transitions across mobile and desktop.
- Expanded the experience with terminal emulation, procedural art, and embedded mini games.
- Integrated a Gemini-powered chatbot assistant via a server function with Upstash Redis-backed rate limiting.

ADDITIONAL

- **PolyHacks 2025** — Built WildGuard, an AI ecological monitoring system using YOLOv8 for real-time wildlife recognition on live video.
- **Private Tutor** — Mathematics, Physics & CS for college/university students (2021 – Present).
- **Languages:** English, French, Arabic (fluent) | German (intermediate).
- **Other:** Authorized to work in the U.S. and Canada (dual citizen).